

Scaled Dot-Product Attention

We compute attention as $\text{softmax}(QK^T / \sqrt{d_k}) V$. For large values of d_k , the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients. To counteract this effect, we scale the dot products by $1/\sqrt{d_k}$.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

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Positional encodings inject order information into the sequence.